

WEEK 6

HEALTHCARE VISUALIZATION AND DASHBOARD

Introduction :

In Week 6, the healthcare dataset was analyzed using Power BI. The main focus of this week was to create DAX measures and develop an interactive healthcare dashboard.

Five important measures were created to calculate patient count, billing amount, average billing amount, average length of stay, and readmission rate. These measures were used to create different visualizations and analyze the healthcare data in a simple and understandable way.

The final dashboard was designed using two pages. Each page presents different aspects of the healthcare data, while the slicers provide interactive filtering across the dashboard.

Objectives :

- To create DAX measures for healthcare data analysis.
- To calculate important patient and billing metrics.
- To create an interactive healthcare dashboard.
- To analyze patient and healthcare-related information using visualizations.
- To compare billing, admission, medical condition, and patient-related information.
- To use slicers for interactive filtering.
- To present the analysis in a clear and organized dashboard.

DAX Measures :

Five DAX measures were created for the healthcare analysis.

Total Patient :

Total Patient = `COUNTROWS(Billing_Table)`

This measure calculates the total number of patient records available in the dataset.

Total Billing Amount :

Total Billing Amount = `SUM(Billing_Table[Billing Amount])`

This measure calculates the total billing amount from the healthcare records.

Average Billing Amount :

Average Billing Amount = `AVERAGE(Billing_Table[Billing Amount])`

This measure calculates the average billing amount for the available records.

Average Length of Stay :

Average Length of Stay =
`AVERAGEX(Billing_Table,DATEDIFF(Billing_Table[Date of Admission],Billing_Table[Discharge Date],DAY))`

This measure calculates the average number of days patients stayed in the hospital.

Readmission Rate :

Readmission Rate =
`VAR TotalCount = [Total Patient]`

`VAR RepeatPatients =
COUNTROWS(
FILTER(
ADDCOLUMNS(
VALUES(Billing_Table[Patient_ID]),`

```
        "AdmissionCount",  
        CALCULATE(COUNTROWS(Billing_Table))  
    ),  
    [AdmissionCount] > 1  
    )  
    )  
  
    RETURN  
    DIVIDE(RepeatPatients, TotalCount, 0)
```

This measure identifies patients with more than one admission and calculates the readmission rate.

Healthcare Visualization & Dashboard :

The healthcare dashboard was created using the DAX measures and different Power BI visualizations. The dashboard contains two pages. The visualizations are arranged separately on each page to make the analysis easier to understand.

Page 1 – Healthcare Dashboard :

Page 1 presents the main summary and comparison visuals of the healthcare data.

Summary Cards:

Four Cards were created to display the important healthcare metrics:

- Total Patient
- Total Billing Amount
- Average Billing Amount
- Average Length of Stay

These cards provide a quick overview of the major values and help users understand the overall healthcare data easily.

Patient Distribution by Age Group :

A Pie Chart was used to show the distribution of patient records based on test results.

The chart represents categories such as:

- Normal
- Abnormal
- Inconclusive

This visualization provides an overall view of the different test result categories in the dataset.

Healthcare Analysis Matrix :

A Matrix was created to provide a detailed comparison of patient and billing information.

The Matrix uses:

- Rows: Age Group, Gender
- Columns: Medical Condition
- Values: Total Patient, Average Billing Amount

This visual helps compare patient records and average billing amounts across different age groups, genders, and medical conditions.

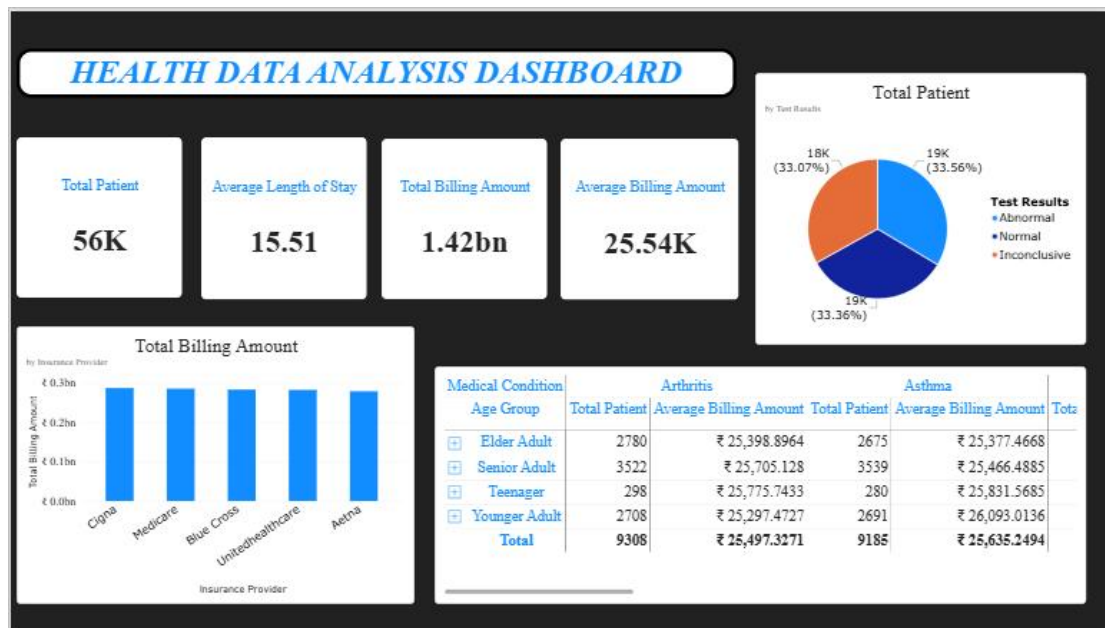
Billing Amount by Insurance Provider :

A Clustered Column Chart was created to compare the total billing amount across different insurance providers.

The visual uses:

- X-axis: Insurance Provider
- Y-axis: Total Billing Amount

This chart provides a clear comparison of billing amounts associated with different insurance providers.



Page 2 – Detailed Healthcare Analysis :

Page 2 contains additional visualizations that provide more detailed analysis of patient, admission, billing, and healthcare-related information.

Total Patient and Total Billing Amount :

A Line and Clustered Column Chart was created to compare total patient records and total billing amount over time.

The visual uses:

- X-axis: Date of Admission
- Column Y-axis: Total Patient
- Line Y-axis: Total Billing Amount

This visualization helps understand changes in patient records and billing amounts across the admission period.

Patient Analysis by Medical Condition and Admission Type :

A Stacked Bar Chart was created to compare patient records across different medical conditions and admission types.

The visual uses:

- Y-axis: Medical Condition
- X-axis: Total Patient

- Legend: Admission Type

This chart helps compare patient records for different medical conditions and shows the contribution of different admission types.

Test Results Analysis :

A Donut Chart was used to display the distribution of patients across different age groups.

The visual uses:

- Legend: Age Group
- Values: Total Patient

This visualization helps understand how patient records are distributed across different age categories.

Average Length of Stay and Average Billing Amount :

A Scatter Chart was created to compare average length of stay with average billing amount.

The visual uses:

- X-axis: Average Length of Stay
- Y-axis: Average Billing Amount
- Legend: Medical Condition
- Tooltips: Total Patient

This visualization helps compare medical conditions based on their average length of stay and average billing amount.

Slicers and Interactive Filtering :

Slicers were added to make the dashboard interactive. The slicers are applicable to both Page 1 and Page 2. When a selection is made, the values in the relevant dashboard visuals are updated accordingly.

Three slicers were used:

Insurance Provider :

The Insurance Provider slicer allows users to filter the dashboard based on the selected insurance provider.

Admission Type :

The Admission Type slicer allows users to filter the dashboard based on different admission types.

Date of Admission :

The Date of Admission slicer allows users to select a specific date range and analyze the corresponding healthcare information.

The slicer settings and selections are displayed on Page 2, while the filtering effect can be observed across both dashboard pages.



Dashboard Analysis :

The dashboard provides different views of the healthcare data.

Page 1 gives a quick overview using summary cards, age group distribution, a detailed Matrix, and insurance provider billing comparison.

Page 2 provides additional analysis of patient count, billing amount, medical conditions, admission types, test results, and the relationship between length of stay and billing amount.

The slicers make the dashboard interactive and allow users to filter the information based on insurance provider, admission type, and

admission date. The selected filters affect the relevant values displayed on both pages.

Conclusion :

Week 6 focused on creating DAX measures and developing an interactive healthcare dashboard using Power BI. Five measures were created to calculate important healthcare metrics such as total patients, total billing amount, average billing amount, average length of stay, and readmission rate.

The final dashboard was organized into two pages with different visualizations for summary and detailed analysis. Cards, Donut Charts, Matrix, Clustered Column Chart, Stacked Bar Chart, Line and Clustered Column Chart, and Scatter Chart were used to present the healthcare information.

Slicers were also added to make both dashboard pages interactive. Overall, Week 6 provided practical experience in creating DAX measures, designing Power BI dashboards, using visualizations, and performing interactive healthcare data analysis.